

- Q.23 Differentiate between absolute and incremental coordinate system.
- Q.24 Write any five advantages of electric actuators.
- Q.25 Write applications of Tachometer.
- Q.26 Name types of sensors. Explain any one.
- Q.27 Distinguish between G-Code and M-Code.
- Q.28 Explain AGV's & its applications.
- Q.29 Explain any two common problems in mechanical components of CNC machine tools.
- Q.30 Write any five advantages of automation.
- Q.31 Explain FMS in detail.
- Q.32 Enlist the various types of joints in robot.
- Q.33 Explain with examples the operation of canned cycle.
- Q.34 Explain the principle and working of LVDT with neat sketch.
- Q.35 Explain various types of tool magazines.

#### SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Define MCU. Explain different types of MCU's.
- Q.37 How will you classify CNC machine faults?
- Q.38 Define Automation. Explain the different types of automation.

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Time : 3Hrs.

M.M. : 100

#### SECTION-A

**Note:** Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 DNC stands for
- Direction & number control
  - Direct Numerical Control
  - Defined Number Control
  - Direct Number code
- Q.2 Full form of FMS:
- Force manufacturing system
  - Free Manufacturing System
  - Formal manufacturing system
  - Flexible Manufacturing system
- Q.3 \_\_\_\_\_ is a device capable of converting energy from one form into another.
- Sensor
  - Micrometer
  - Transducer
  - Actuator

Q.4 \_\_\_\_\_ is an instrument used to measure the rotation speed of an object?

- a) Transducer                      b) Thermometer
- c) Dynamometer                  d) Tachometer

Q.5 Full form of AGV is :

- a) Absolute Guided Vehicle
- b) Armed Guided Vehicle
- c) Automatic Guided Vehicle
- d) Actuated Guided Vehicle

Q.6 \_\_\_\_\_ is based on see back's effect.

- a) Thermometer                  b) Thermocouple
- c) Thermistor                    d) None of these

Q.7 A systematic layout to accommodate the required number of tools is called \_\_\_\_\_.

- a) Tool Magazine                b) Tool layout
- c) Tool holder                    d) None of these

Q.8 Open loop control system is also known as \_\_\_\_\_ control system.

- a) Feedback                      b) Non feedback
- c) Hydraulic                      d) Pneumatic

Q.9 Full form CAM is :

- a) Computer Aided Manufacturing
- b) Central Aided Manufacturing
- c) Common Aided Manufacturing
- d) Cyclic Aided Manufacturing

Q.10 Incremental system is also known as

- a) Fixed Zero                      b) Floating zero
- c) Either of (a) or (b)          d) None of these

### SECTION-B

**Note:** Objective type questions. All questions are compulsory. (10x1=10)

Define the following :

Q.11 Part program.

Q.12 Magnetic tape.

Q.13 Qualified Tools.

Q.14 Actuator.

Q.15 Passive transducer.

Q.16 Swarf Removal.

Q.17 M-Code.

Q.18 Flexible Automation.

Q.19 Robot.

Q.20 CAD.

### SECTION-C

**Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Write any five advantages of NC system.

Q.22 Name types of Adaptive Control System. Explain any one.